# MSG No. 113.272 $P_c\bar{4}2_1m$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `-4..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,4,5,6]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[2,3,7,8]` |
| 3 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[9,12,13,14]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[10,11,15,16]` |

Table 2: Wyckoff site: `4b`, site symmetry: `-4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{4}\right]$ | `[1,4,13,14]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[2,3,15,16]` |
| 3 | $\left[0,\ 0,\ \frac{3}{4}\right]$ | `[5,6,9,12]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[7,8,10,11]` |

Table 3: Wyckoff site: `4c`, site symmetry: `2.mm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,4,7,8]` |
| 2 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | `[2,3,5,6]` |
| 3 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9,12,15,16]` |
| 4 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$ | `[10,11,13,14]` |

Table 4: Wyckoff site: `8d`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | `[2,3]` |
| 3 | $[0,\ 0,\ -z]$ | `[5,6]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | `[7,8]` |
| 5 | $\left[0,\ 0,\ z+\frac{1}{2}\right]$ | `[9,12]` |
| 6 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[10,11]` |
| 7 | $\left[0,\ 0,\ \frac{1}{2}-z\right]$ | `[13,14]` |
| 8 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[15,16]` |

Table 5: Wyckoff site: `8e`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ x+\frac{1}{2},\ z\right]$ | `[1,8]` |
| 2 | $\left[x+\frac{1}{2},\ -x,\ -z\right]$ | `[2,5]` |
| 3 | $\left[\frac{1}{2}-x,\ x,\ -z\right]$ | `[3,6]` |
| 4 | $\left[-x,\ \frac{1}{2}-x,\ z\right]$ | `[4,7]` |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 5 | $\left[x,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [9,16] |
| 6 | $\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$ | [10,13] |
| 7 | $\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}-z\right]$ | [11,14] |
| 8 | $\left[-x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [12,15] |

Table 6: Wyckoff site: 16f, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | [1] |
| 2 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$ | [2] |
| 3 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$ | [3] |
| 4 | $[-x,\, -y,\, z]$ | [4] |
| 5 | $[y,\, -x,\, -z]$ | [5] |
| 6 | $[-y,\, x,\, -z]$ | [6] |
| 7 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$ | [7] |
| 8 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$ | [8] |
| 9 | $\left[x,\, y,\, z+\frac{1}{2}\right]$ | [9] |
| 10 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [10] |
| 11 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [11] |
| 12 | $\left[-x,\, -y,\, z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[y,\, -x,\, \frac{1}{2}-z\right]$ | [13] |
| 14 | $\left[-y,\, x,\, \frac{1}{2}-z\right]$ | [14] |
| 15 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [15] |
| 16 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [16] |